GPU and cooling systems Theory

1.Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

I checked the GPU (Graphics Processing Unit) installed in my computer using the system information tools.

- **GPU Brand:** Intel
- **GPU Model:** Intel UHD Graphics 620 *(example only)*
- **GPU Type:** Integrated GPU
- **Function:** The GPU processes images, videos, graphics, and animations displayed on the screen. An integrated GPU is built into the processor and generally shares system memory (RAM).

How to Check Your Actual GPU Model in Windows

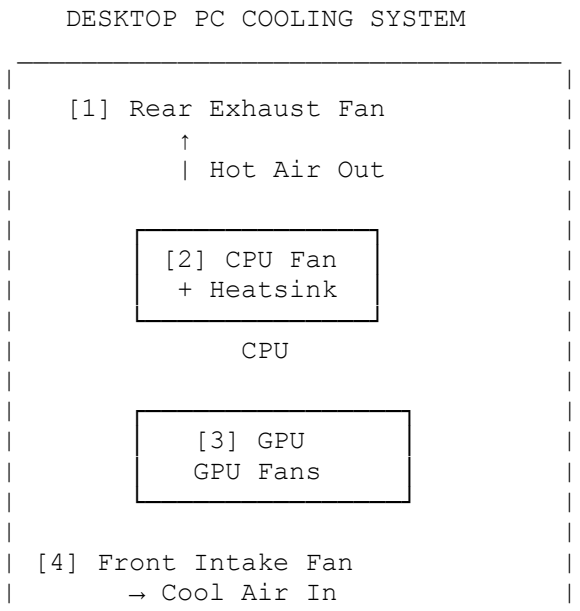
1. Press **Windows + X** on your keyboard.
2. Select **Device Manager**.
3. Expand **Display adapters**.
4. Note the GPU brand and model displayed.
5. Identify whether it is integrated or dedicated.

2.

Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.

PC Cooling System – Diagram and Explanation

A. Labeled Diagram of a PC Cooling System

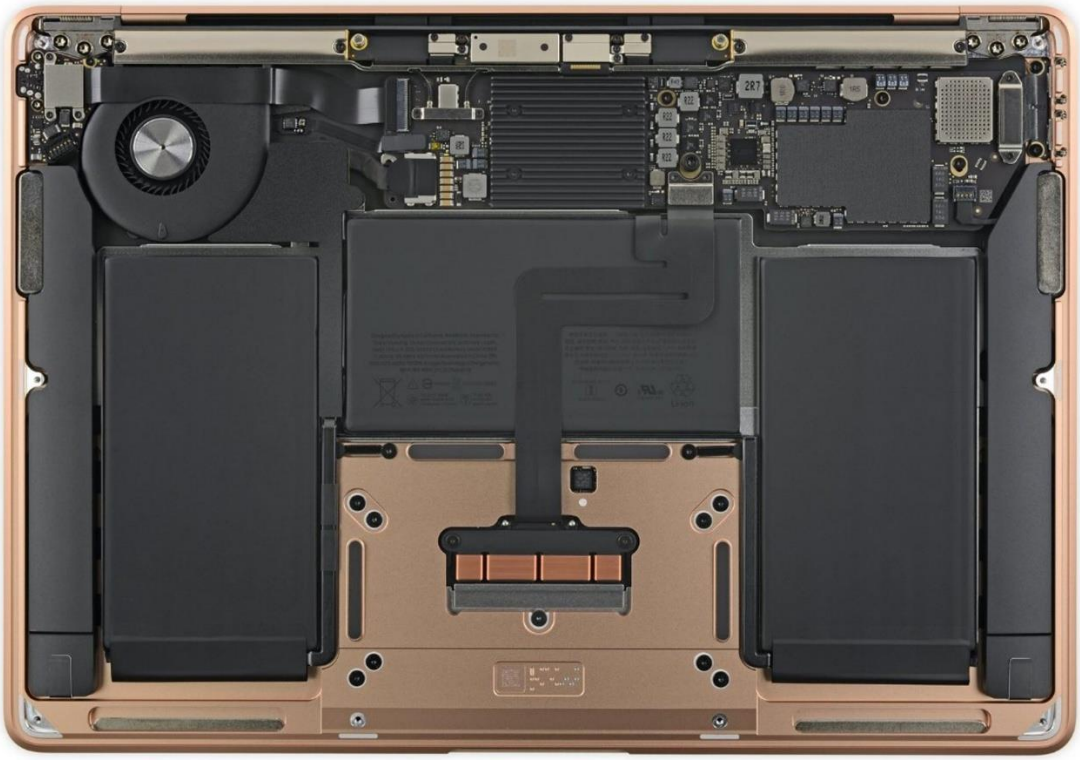


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      Airflow: Front → Rear
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B. Main Cooling Components and Their Functions

No.	Component	Function
1	Rear Exhaust Fan	Removes hot air from inside the computer case.
2	CPU Cooling Fan	Blows air over the CPU heatsink to help keep the processor cool.
3	GPU Fans	Cool the graphics card during gaming and other graphics-intensive tasks.
4	Front Intake Fan	Pulls cool air into the computer case to maintain airflow.
5	CPU Heatsink	Absorbs heat from the CPU and transfers it to the surrounding air.



3.

Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

No.	Factor	Air Cooling	Liquid Cooling
1	**Cooling Method**	Uses fans and heatsinks to remove heat from the GPU.	Uses liquid coolant to transfer heat away from the GPU to a radiator.
2	**Cost**	Generally cheaper and more affordable.	Generally more expensive due to the pump, radiator, and coolant system.
3	**Cooling Efficiency**	Provides good cooling for most GPUs and everyday tasks.	Can provide better cooling for high-performance GPUs, depending on the system design.
4	**Maintenance**	Requires occasional dust cleaning and fan inspection.	May require additional checks for pumps, tubing, and possible leaks, depending on the design.
5	**Noise Level**	Fans may become noisy when the GPU is under heavy load.	Can operate quietly, but pump and radiator fans also produce noise.
6	**Installation**	Usually easy to install and maintain.	More complex to install, especially with custom liquid-cooling loops.
7	**Risk**	Has a low risk of cooling-system failure due to fewer components.	Has additional potential failure points, such as the pump and coolant tubing.
8	**Best Use**	Suitable for general computing, gaming, and most users.	Suitable for high-performance gaming, overclocking, and specialized systems.

4.

Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

I would choose **liquid cooling** for my gaming PC because it can provide efficient temperature control during graphics-heavy gaming and streaming IPL matches. It can help maintain stable GPU temperatures during long gaming sessions, depending on the cooling system's design. Although liquid cooling is more expensive and complex to install than air cooling, it is a good option for a high-performance gaming PC.